Do Now · retrieval

Lesson 065 · Computational Methods: Recognising Solvable Problems

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a precondition?

Q2

What is abstraction?

Q3

How does a queue differ from a stack?

Q4

Trace binary search for 11 in the sorted, zero-indexed array [2,4,6,8,10,12,14]. Start low=0, high=6. Use mid=(low+high) DIV 2 and exclude the checked midpoint when updating bounds. State the checked indexes and the final result.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify — A solvable computational problem needs an adequately precise input representation, a finite method and a verifiable output or objective.

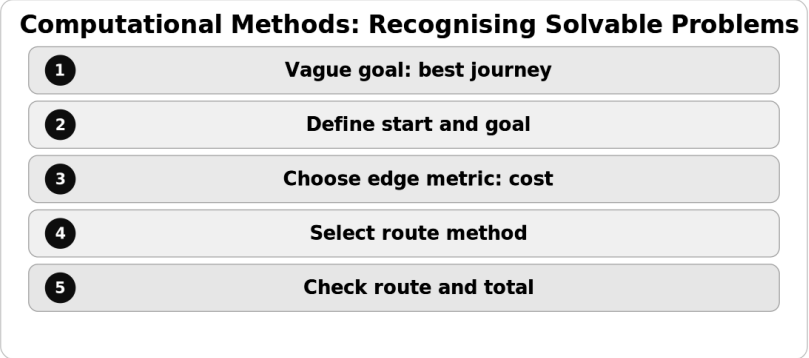
Core idea 2

Model — Problem recognition identifies the class of task—classification, searching, route finding, scheduling or optimisation—before selecting a method.

Core idea 3

Evaluate — Define measurable success criteria and constraints; vague goals such as 'best route' require clarification of 'best' (distance, cost or time).

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

problem recognition — Identify the underlying task type.

constraint — Requirement a solution must obey.

objective function — Quantity to minimise or maximise.

computable specification — Precise task that a program can implement/check.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Turn 'choose the best bus route' into a precise computational problem.

Follow the worked method

1. Explain the decision — Cost is defined and connections must be valid; output stops and total fare.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the decision

An ordering rule makes fairness measurable and repeatable within the chosen model.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define computational problem input, process, output for cheapest route on a network. [6]

Question 2

A pupil asks to 'sort books by best'. Give two precise orderings, explain why different and select one for finding a title. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate whether a computer can choose a 'good teacher' simply from exam results, explaining models, constraints and limits. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Every vague human objective has exactly one obvious numerical optimisation target.

Use key vocabulary from memory

1. Explain problem recognition in your own words.

2. Explain constraint in your own words.

3. Explain objective function in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.